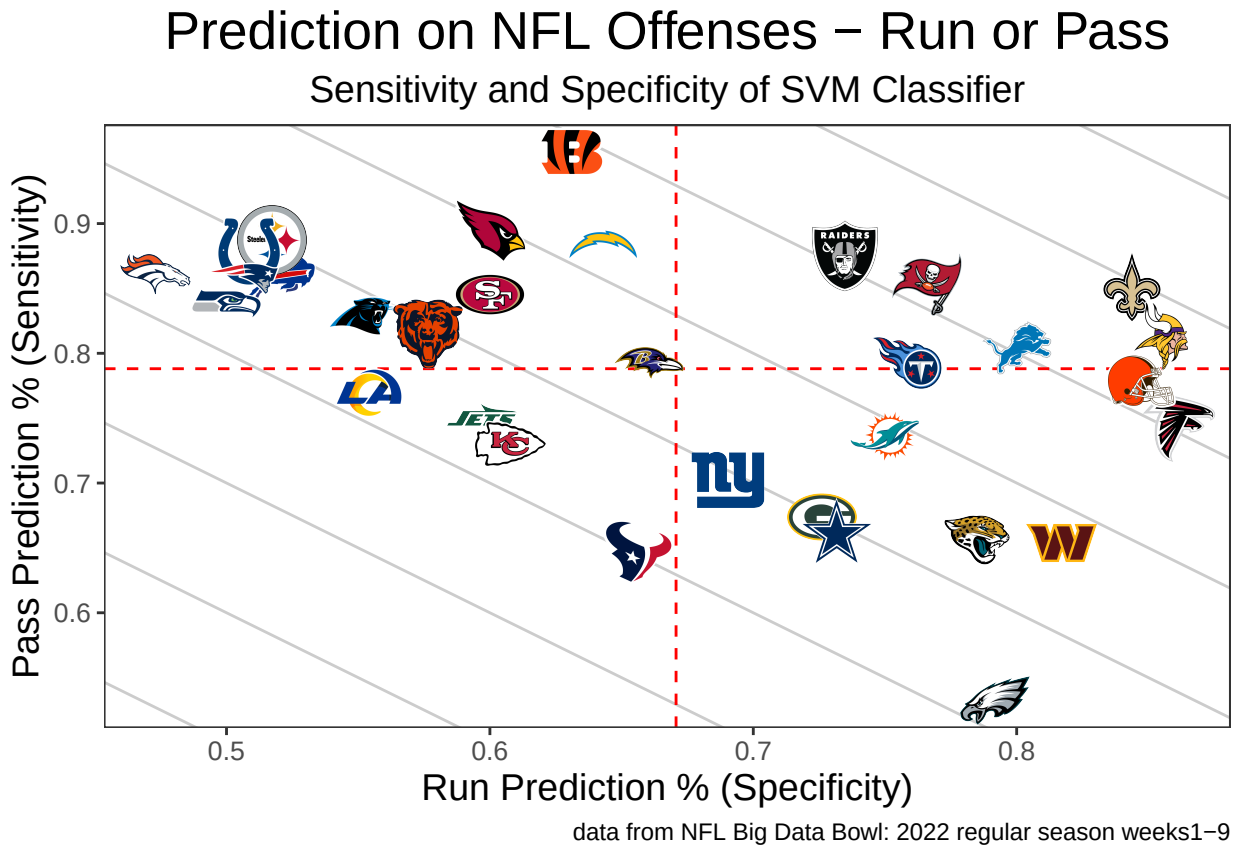
# NFL Graphs - C

Chandler

2024-12-03

Positive class = "pass" Negative class = "run" Sensitivity refers to pass Specificity refers to run

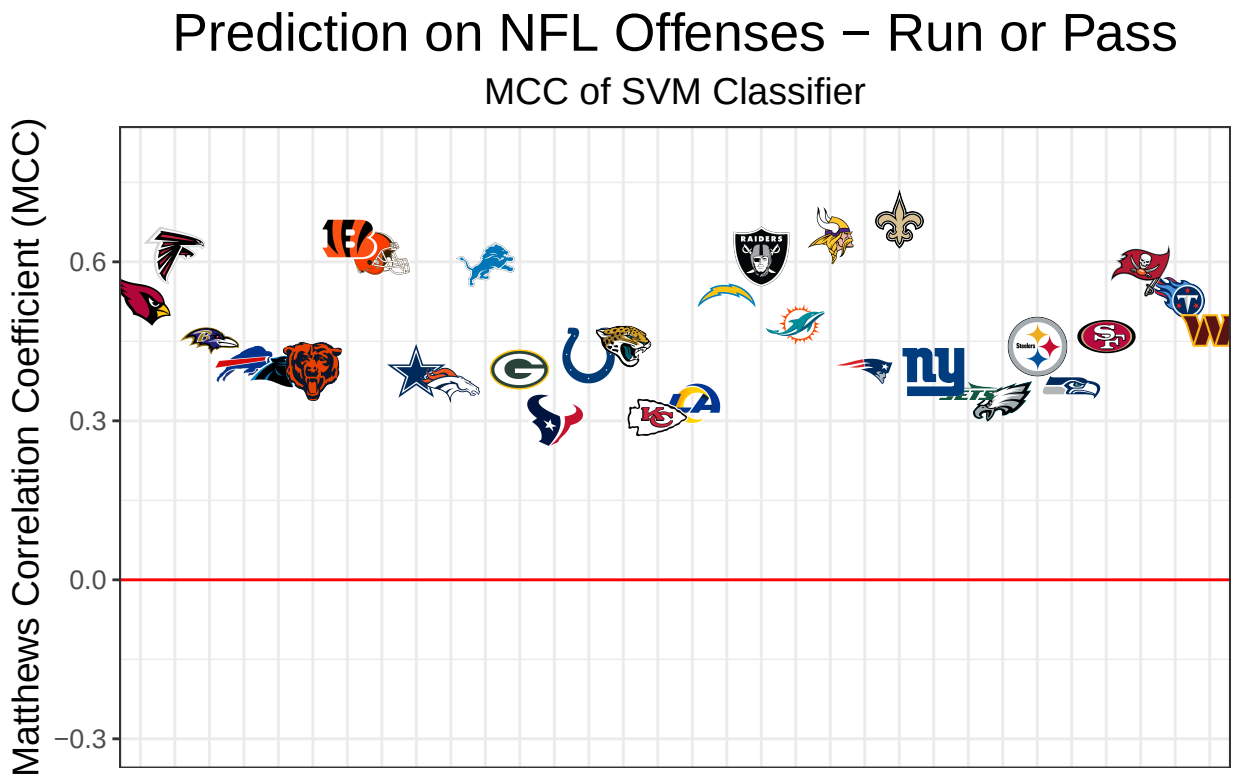
```r
# this number will scale the image sizes proportionally
s <- 1.25
spec_sens_svm <- svm_data %>%
  ggplot(aes(y = sens, x = spec)) +
  # draw the background lines. sequence for intercept will need changing
  ggplot2::geom_abline(slope = -1, intercept = seq(2, 0.6, -0.1), alpha = .2) +
  # draw the red dashed mean lines for the data
  nflplotR::geom_mean_lines(aes(x0 = spec , y0 = sens)) +
  geom_nfl_logos(aes(team_abbr = team, width = 0.075 * s, height = 0.1 * s)) +
  labs(title = "Prediction on NFL Offenses - Run or Pass",
       subtitle = "Sensitivity and Specificity of SVM Classifier",
       y = "Pass Prediction % (Sensitivity)",
       x = "Run Prediction % (Specificity)",
       caption = "data from NFL Big Data Bowl: 2022 regular season weeks1-9") +
  theme_bw() +
  theme(
    # this just makes the background flat white
    plot.background = element_rect(fill = "white"),
    panel.grid = element_blank(),
    plot.title = element_text(hjust = 0.5, size = 20),
    plot.subtitle = element_text(hjust = 0.5, size = 14),
    axis.text = element_text(size = 10),
    axis.title = element_text(size = 14)
  )
spec_sens_svm
```




```r
s <- 2
spec_sens_svm_by_winloss <- svm_data %>%
  ggplot(aes(y = sens, x = spec)) +
  facet_wrap(wining_record~., nrow = 2) +
  geom_nfl_logos(aes(team_abbr = team, width = 0.075 * s, height = 0.1 * s)) +
  labs(title = "SVM Classifier Prediction on NFL Offenses - Run or Pass",
       subtitle = "2022 Regular Season Through Week 9",
       y = "Pass Prediction % (Sensitivity)",
       x = "Run Prediction % (Specificity)",
       caption = "data from NFL Big Data Bowl") +
  theme_bw()
```

```r
s <- 1
mcc_svm <- svm_data %>%
  ggplot(aes(y = mcc, x = team)) +
  geom_nfl_logos(aes(team_abbr = team, width = 0.075 * s, height = 0.1 * s)) +
  geom_hline(yintercept = 0, color = "red") +
  labs(title = "Prediction on NFL Offenses - Run or Pass",
       subtitle = "MCC of SVM Classifier",
       y = "Matthews Correlation Coefficient (MCC)",
       x = "",
       caption = "data from NFL Big Data Bowl: 2022 regular season weeks1-9") +
  scale_y_continuous(limits = c(-0.3, 0.8)) +
  theme_bw() +
  theme(axis.ticks.x = element_blank(),
        axis.text.x = element_blank(),
```

```
        plot.title = element_text(hjust = 0.5, size = 20),
        plot.subtitle = element_text(hjust = 0.5, size = 14),
        axis.text.y = element_text(size = 10),
        axis.title.y = element_text(size = 14))
mcc_svm
```




```
svm_data %>% rename(svm_mcc = 'mcc') %>%
  select(team, wins, svm_mcc) %>%
   left_join(log_data %>% select(team, log_mcc),
                              by = "team") %>%
  arrange(svm_mcc) %>%
  slice(1:5) %>%
  kable()
```

| team | wins | svm_mcc | log_mcc |
|---|---|---|---|
| HOU | 1 | 0.3004505 | 0.3042903 |
| KC | 6 | 0.3065196 | 0.1307441 |
| LA | 3 | 0.3319766 | -0.0091264 |
| PHI | 8 | 0.3371568 | 0.1646828 |
| NYJ | 6 | 0.3509050 | 0.1124527 |